Mercantail: Pet & Merge Prototype

Playtest & Product Brief

Format: Prototype · 10 sessions · 5,393 events logged · ~114 minutes total play time · March 2026

Github: https://github.com/ds30mins/mercantail-pet-merge-prototype · By Diana Sinohardjo

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1. What I Built & Why

Mercantail is a mobile-first Merge-2 prototype built around one question: does the commerce loop hold attention before the meta layer gives players a reason to return?

Merge games live or die on session clarity. Before layering in pet evolution and progression, I needed to know if the core loop — order targets, energy pacing, reward timing — could hold attention on its own. So I built a playable prototype and tested it with 10 players across 5,000+ events. This brief is what I found.

2. Playtest Methodology

Parameter	Detail
Sessions	n=10 across 8 players (P01–P08)
Format	Mix of remote self-recorded, observed, and embedded in-game logger
Event logging	5,393 events across 17 data dimensions
Metrics tracked	Session length, spawn/merge/order/shop/energy events, tier progression, churn signals, gold economy
Logger upgrade	P08's sessions used an embedded in-game logger built into the prototype — capturing item paths, decimal timestamps, and full economy state automatically. Earlier sessions used manual event logging.

Compensation	None. Players were casual mobile gamers with no prior merge-game briefing.
Note	Sample size is small and convenience-sampled. Findings are directional, not statistically conclusive. Sufficient to form hypotheses for formal A/B testing.

Prototype scope note: The full Mercantail concept includes pet evolution and park building as the long-term retention layer. The prototype deliberately excludes these — by design. If players disengage during the core merge-and-fulfill loop, adding a meta layer won't fix it. So the call was to validate the session-level mechanic first. This playtest is that validation step. The 100% order fulfillment rate and sustained engagement in longer sessions are the green light to invest in building the pet evolution arc.

> Data design note: The 17 dimensions were chosen to capture the full session loop — spawn, merge, order, shop, and energy events — plus economy state (Gold before/after per event) and qualitative signals (churn, shop voluntariness). Early sessions used manual logging from video review, which introduced two quality limits: no decimal timestamps and no automatic item path tracking. This gap was a design decision trigger. For P08's sessions (S08, S10), I built an embedded in-game logger directly into the prototype to capture complete economy state and item paths automatically. The resulting split — Basic Data (S01–S07, S09) vs Rich Data (S08, S10) — is treated as a methodology variable throughout this analysis: tier funnel and economy calculations use Rich Data only, where the log is complete.

3. Session Summary Data

All numbers derived from event-level log (5,393 rows). Sessions S08 and S10 (highlighted) use Rich Data from the embedded logger. All other sessions use Basic Data from manual event logging.

Sess.	Player	Length	Events	Spawned	Merged	Orders	Energy	Ratio	Max T	Data
S01	P01	2m 5s	63	34	16	8	2	0.47	3*	Basic
S02	P02	3m 14s	64	34	20	3	2	0.59	4*	Basic
S03	P03	0m 49s	16	13	0	2	0	0.00	1*	Basic
S04	P04	8m 34s	414	180	123	47	25	0.68	4*	Basic
S05	P05	14m 47s	632	289	235	37	32	0.81	5*	Basic
S06	P06	9m 26s	453	224	149	39	17	0.67	5*	Basic
S07	P07	14m 59s	520	247	179	39	21	0.72	5*	Basic
S08	P08	30m 29s	1,560	705	560	128	69	0.79	5**	Rich
S09	P02	2m 32s	68	30	26	4	3	0.87	5*	Basic
S10	P08	27m 27s	1,601	719	562	144	70	0.78	5**	Rich

** Max tier confirmed via video review (manual sessions did not capture tier in event log). ** Embedded in-game logger. P02, P08 were observed live. Others were remote and provided self-recorded video.*

4. Player Funnel Analysis

All numbers derived from event-level log (5,393 rows).

Funnel Stage	Count	%	Benchmark	Implication
Started session	10/10	100%	100%	Baseline
Reached Tier 2+ item	9/10	90%	No public benchmark	Core loop discoverability strong once players engage
Reached Tier 5 (max)	6/10	60%	No public benchmark	Significant depth engagement, but no Tier 5 orders creates a dead end
Fulfilled ≥1 order	10/10	100%	No public benchmark — competitively sensitive	Goal clarity is working across all sessions.
Opened shop	9/10	90%	No public benchmark	Energy gate reliably drives shop engagement
Used energy refill	9/10	90%	No public benchmark	Monetisation moment validated across nearly all sessions
Session >3 minutes	7/10	70%	70% crossed 3.1–3.5 min median (GameAnalytics 2025)	Players who find the loop stay significantly longer

5. Key Findings

Finding 1: Order fulfillment is the strongest engagement signal

Every session (10/10, 100%) included at least one fulfilled order. Players in S04–S07 fulfilled 37–47 orders each. P08's two sessions produced 128 and 144 orders respectively — the highest of any tester. The commerce loop is intrinsically motivating once players understand it.

No published industry benchmark exists for first-session order fulfillment — this metric is treated as competitively sensitive across the industry. The 100% rate stands on its own signal.

> *"Core loop-nya oke, sederhana dan mudah dipelajari."* — P04
>
> *"I don't think I ever rate a game... 5 stars."* — P01

Finding 2: Tier progression is strong — but Tier 5 has no purpose

Tier 2+ was reached in 9 of 10 sessions (90%), exceeding the ~80% benchmark. 6 of 10 sessions reached Tier 5 (the maximum). However, this surfaces a new and significant design gap: there are no Tier 5 orders. Players who invested in reaching the maximum evolution tier found no reward for doing so. P08 raised this directly and independently.

The merge funnel quantifies why this matters. Using Rich Data sessions only (S08 + S10), items produced at each tier are compared against the 2:1 theoretical baseline — since two items of tier N must be merged to produce one item at tier N+1:

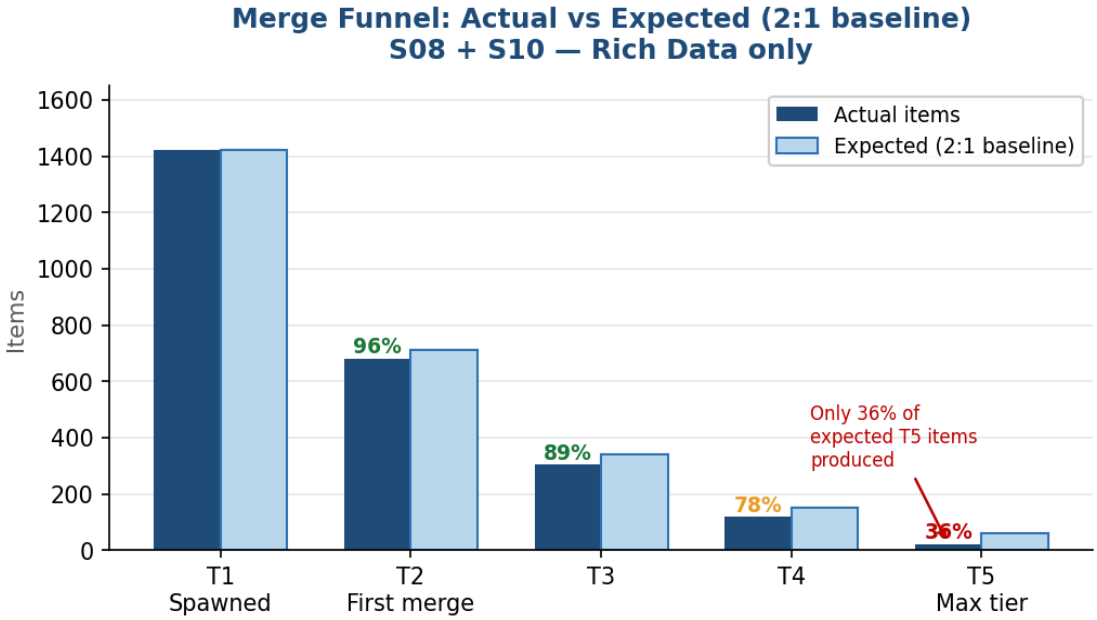

Fig 1. Merge funnel — actual vs expected items per tier. S08 + S10 (Rich Data only). Efficiency % shown above each actual bar.

T1 through T3 efficiency is strong — players merge actively at 96% and 89% of theoretical baseline. The drop begins at T4 (78%) and collapses at T5: only 21 T5 items were produced against an expected 59 — a 36% efficiency rate. This is not random attrition. With no Tier 5 orders to justify the ~50-spawn investment required to produce a single T5 item, players rationally consume their T4 items for lower-tier orders instead. The dead end is confirmed in the data.

> *"Why no order Lv5? What's the purpose of Lv5 item?"* — P08

Finding 3: Session length is bimodal — a discovery threshold exists

Session lengths split into two distinct clusters: short (49–194s) and long (514–1,829s). There is no middle ground. Players either find the loop engaging and stay for extended sessions, or they disengage before it clicks.

The median mobile game session runs 3.1–3.5 minutes (GameAnalytics 2025). 70% of Mercantail sessions crossed that threshold. For players who broke through, sessions ran well above it — the long cluster averaged over 8 minutes, placing those sessions in the top 10% of industry session lengths. P08's two sessions at 30 and 27 minutes sit in the top 1% — territory GameAnalytics associates with games that have deep progression, social mechanics, or content that encourages extended uninterrupted play.

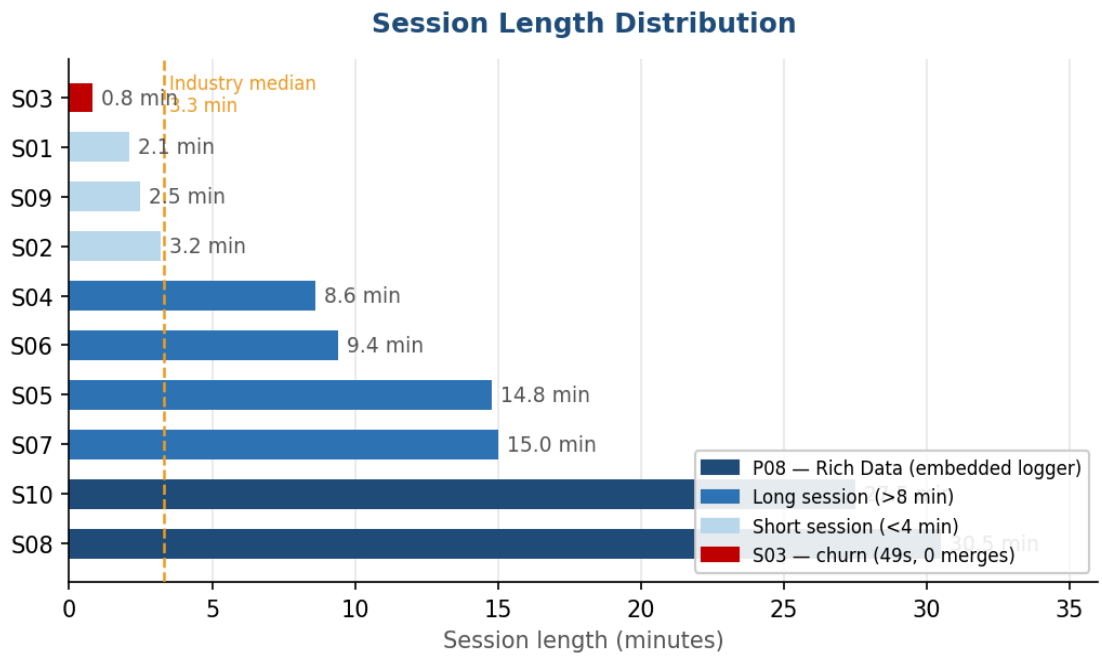

Fig 2. Session length distribution — all 10 sessions. Orange dashed line = industry median (3.3 min). Bimodal split visible: short cluster (S01–S03, S09) vs long cluster (S04–S08, S10).

> *"Kekurangan utama buat gua adalah kurang tujuan atau sense of progression-nya."* — P04

Finding 4: Energy gate drives shop engagement — but emotional context needs attention

Shop open rates tracked energy refill rates almost perfectly across all sessions. 9 of 10 players hit the energy gate and opened the shop. The monetisation moment is reliably triggered. The design question for v2 remains: can the shop also be reached in an aspiration context (wanting more items) rather than always a friction context (blocked from continuing)?

P08's Gold trajectory across both sessions makes the economy calibration signal concrete. Gold is earned at approximately 69 Gold per fulfilled order and spent at 100 Gold per energy refill. The net result:

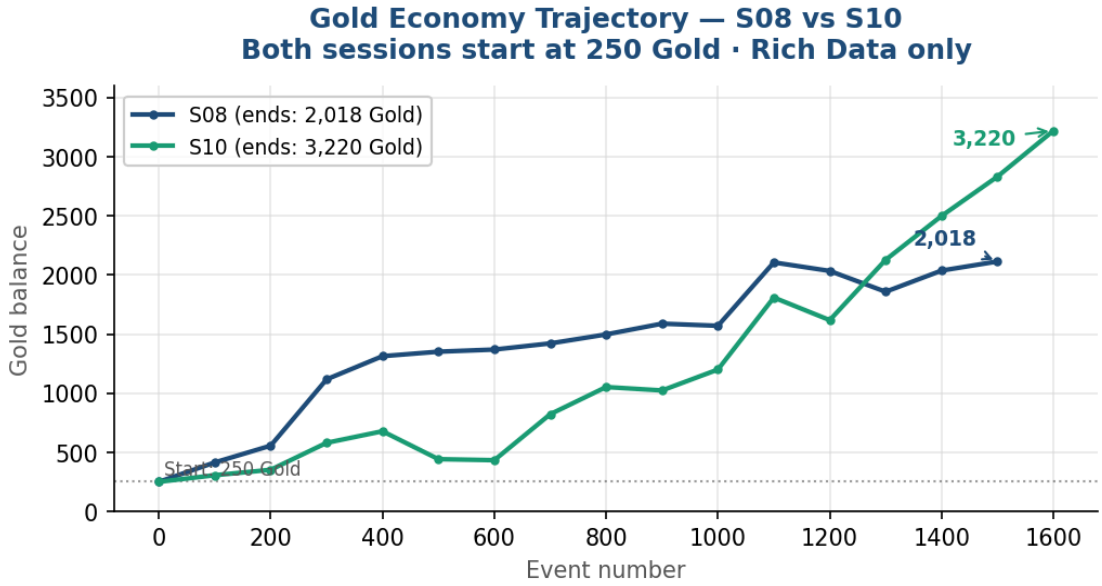

Fig 3. Gold balance over session — S08 vs S10. Both start at 250 Gold. S08 ends at 2,018 (+707%). S10 ends at 3,220 (+1,188%). Each data point = 100 events. Rich Data only.

Gold accumulates consistently across both long sessions — players earn more than they spend. The surplus is not a sign that the economy is healthy: it signals a missing second spend destination. Beyond energy refills, there is currently nothing to spend Gold on. P08 flagged sell prices as too low; the trajectory confirms that the earn-to-spend ratio tilts toward accumulation. Both issues resolve when Tier 5 orders and Pearl-tier items are introduced — which is exactly what the next build phase adds.

> *"Refill energy mungkin bisa ada opsi buat langsung full."* — P05
>
> *"Sell price is too low."* — P08

Finding 5: Four independent players converged on the same missing layer — progression & stakes

P03, P04, P07, and P08 — with no contact with each other — independently raised the same gap: the game lacks escalating challenge, a win/lose condition, checkpoint system, or a sense of meaningful progression beyond gold accumulation. P08 specifically asked about endings, levels, and checkpoints. This convergent unprompted feedback across four players is the strongest signal in the dataset.

> *"Ga ada game over atau timer-nya? Kalo ada timer bakal better."* — P03
>
> *"Kurang tujuan... selain uang, ga ada lagi ukurannya."* — P04
>
> *"How's the ending? Need checkpoint / level / progress."* — P08

Finding 6: Tutorial is the clearest onboarding gap

Three players (P03, P07, P08) independently requested a tutorial or guided onboarding. P08's feedback was direct: the game needs a tutorial. This is consistent with the S03 churn data (49 seconds, Tier 1 only, no merges completed) — the free-play start is not sufficient to orient new players.

> *"Tutorial ditaruh di pojok kanan atas agak kurang membantu."* — P03
> *"Perlu ada help interactive di awal buat new player."* — P07
> *"Need tutorial."* — P08

Finding 7: UI legibility — icons and level indicators too small

P08 noted that icons and level numbers within item tiles are too small on a mobile screen. This is a usability finding rather than a design finding — a straightforward UX fix that would reduce cognitive load during play, particularly for new players trying to understand the evolution tree.

Finding 8: Data quality note — shop_opened events may be inflated

A touch sensitivity issue in the prototype means accidental shop closes may have triggered additional "shop_opened" events. "Energy_refill" count is the cleaner signal for monetisation intent, refills require deliberate confirmation. Recommended fix before next playtest round.

6. Action Plan for Key Findings

Finding	Signal	Action Plan
Finding 1: Order fulfillment — 100% rate	Core loop is intrinsically motivating. Green light to build the meta layer.	Validates concept — no action needed
Finding 2: Tier 5 has no purpose	Players who invest in reaching max tier find no reward. Unambiguous design gap.	Implement next
Finding 3: Session length is bimodal	Sessions split: under 3 minutes or over 8 minutes. Onboarding is the lever.	Test next (onboarding)
Finding 4: Shop is friction-only	Monetisation moment reliably triggered — but always reactive, never aspirational. Sell prices too low.	Implement next
Finding 5: Players demand progression & stakes	Four independent players raised the same gap. Strongest signal in the dataset.	Test next (meta layer)
Finding 6: Tutorial gap	Three independent players requested guided onboarding. S03 churn confirms free-play start is insufficient.	Test next (onboarding)
Finding 7: UI legibility	Icons and level indicators are too small on mobile. Usability fix, not a design question.	Implement next

7. Next Implementation

These are unambiguous fixes. The signal is clear enough that testing would be procrastinating behind data.

Tier 5 orders
Introduce at least one rotating order requiring a Tier 5 item, with a Gold reward that justifies the merge investment (~150–200 Gold, reflecting the ~50-spawn cost to produce a T5 item). Closes the motivation dead end identified by P08. Creates a natural premium spend moment — players who want to accelerate to Tier 5 have a clear reason to refill energy.

Sell price calibration
Increase item sell prices by 25–50% to better align Gold economy with energy refill cost. P08's feedback ("sell price is too low") is consistent with the Gold trajectory data — players accumulate Gold at a rate that makes refill decisions feel punishing rather than meaningful.

Delayed first energy gate
Delay the first energy gate by approximately 50% to allow players to experience the commerce loop before hitting a monetisation moment. Shop opens should have an aspiration context (wanting more), not only a friction context (blocked from continuing).

UI legibility
Increase icon size and level number visibility within item tiles. Reduces cognitive load for new players navigating the evolution tree.

Shop_opened event fix
Fix the touch sensitivity issue where players accidentally close the shop by clicking/tapping anywhere on the screen.

8. A/B Testing Plan

Test 1 — Guided vs. Free-Play Onboarding

Element	Detail
Hypothesis	Players given a 3-step guided merge sequence will reach Tier 2+ at higher rates — without reducing exploration in experienced merge players
Control (A)	Current free-play start — no onboarding direction
Variant (B)	3-step guided merge sequence before free play unlocks
Primary metric	% players reaching Tier 2+ item in first session
Secondary metric	Session length and orders fulfilled in first session
Sample size	~50 sessions per variant
Risk to watch	Guided onboarding may reduce early exploration and feel patronising to experienced merge players

Test 2 — Meta layer direction: pet evolution vs. story progression

Element	Detail
Hypothesis	Pet evolution anchored to order fulfillment will produce stronger D7/D30 retention than a story-gated progression layer — because the reward is immediate and personal rather than narrative and deferred
Control (A)	Pet evolution arc — species hatch, evolve, and become permanently visible in the park as fulfillment milestones are reached
Variant (B)	Story/narrative progression — fulfillment unlocks chapters of a light narrative, giving orders a contextual purpose
Primary metric	D7 and D30 retention
Secondary metric	Session length and orders fulfilled per session in D7 cohort
Sample size	Minimum 100 players per variant — retention tests need time, not just volume
Risk to watch	Pet evolution requires art and animation investment upfront. Story requires a content pipeline. Neither is cheap — validate retention signal at minimum viable scope before full production commitment

Test 3 — Gold refill vs rewarded ad preference

Element	Detail
Hypothesis	Players shown an ad option alongside the Gold refill button will choose ads over Gold at meaningful rates — indicating willingness to engage with rewarded ads as a non-IAP revenue layer
Control (A)	Current shop — Gold refill button only
Variant (B)	Shop with two options — "Refill for 100 Gold" and "Watch ad for 5 min unlimited energy" (simulated with blank screen)
Primary metric	% of shop opens where player chooses ad option vs Gold option
Secondary metric	Session length delta — does the ad option extend sessions?
Sample size	Minimum 100 players per variant
Risk to watch	If players discover the ad option, some Gold-spenders might switch to ads instead — cannibalizing the soft currency spend. Worth measuring Gold refill rate in variant B vs control to check for cannibalization.

9. Monetisation Layer

The prototype validates one monetisation moment: the energy gate. Hard currency, gacha, and order slot expansion are designed but not yet implemented — they belong to the next build phase, not this playtest.

What's validated

Touchpoint	Playtest Evidence	Design Note
Energy refills (soft paywall)	In 9 out of 10 sessions, players hit the energy gate and opened the shop.	Always a friction context — blocked from continuing. First-purchase conversion will likely stay low until the shop can also be reached in an aspiration context. Addressed in Section 7.
Rewarded ads — 10-min unlimited energy	P08 flagged energy running out too fast. P05 requested auto-refill option. The natural response: one rewarded ad per session unlocks 10 minutes of unlimited energy.	Non-IAP revenue layer that serves non-spenders without undermining the soft paywall. Keeps high-engagement players in session longer — directly addressing the P08 archetype.

What's designed, not yet built

Touchpoint	Playtest Evidence	Design Note
Pearl bundles (hard currency)	Primary IAP driver. Used for energy refills, instant order fulfillment, evolution boosts, egg gacha pulls.	Suggested tiers: $0.99 · $4.99 · $9.99. After meta layer direction is confirmed.
Egg gacha	100 Pearls/pull. Common / Rare / Legendary. Legendary at 2% drop rate creates aspirational spend target.	Same phase as Pearl bundles
Order slot expansion	Natural upsell for high-engagement players once order cadence is understood from live data.	Post-launch, data-led

The High-Engagement Archetype — What P08 Showed

Metric	P08 — two sessions combined
Avg session length	~29 minutes
Orders fulfilled	~136 per session
Shop interactions	~87 per session
Energy refills	~70 per session
Gold trajectory	250 → 2,018 → 3,220 across two sessions

P08's Gold trajectory confirms the economy accommodates deep play without breaking — Gold accumulates at a rate that creates meaningful spend decisions rather than arbitrary ones. Whether that translates to real spend depends on what the shop offers beyond energy. The session loop is ready. The spend destination is what we're going to build next.

Assets & Links

- Playable prototype
- GitHub repo
- Portfolio brief (Github)
- Portfolio brief (Google Drive)
- Event log (raw data) (Google Drive)